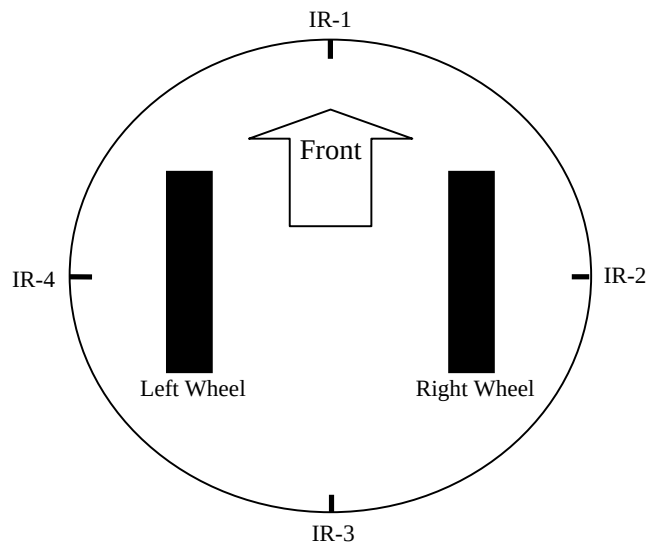


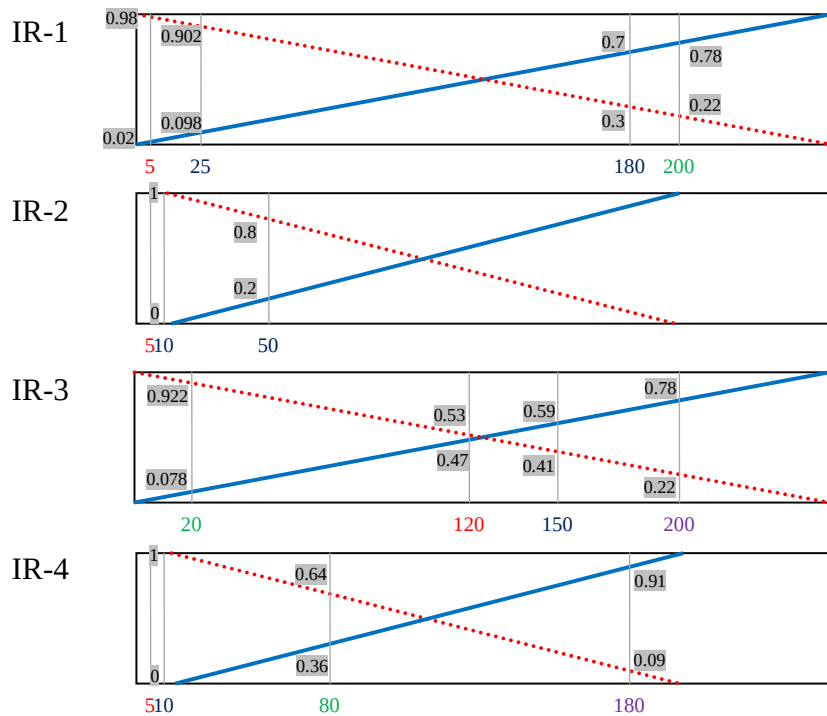
Fuzzy Control Exercise

Let's design a fuzzy control for a very simple robot.

- The robot has only 4 infrared sensors locating around the robot. Each sensor reads the distance to the nearest object in its direction. The possible value is 0 ... 255.
- The robot has two wheels. We can set speed of each wheel from -100 ... 0 ... 100. Positive values mean moving forward, and negative values mean moving backward.



Membership Functions



Rules

	IR-1	IR-2	IR-3	IR-4	TURN	MOVE
1	Near	-	-	-	0	-10
2	-	Near	-	-	-90	-
3	-	-	-	Near	90	-
4	Far	Near	-	Near	0	5
5	Far	Far	-	Far	0	10
6	-	-	Near	-	0	5

Inference

What will be happen to your robot if the robot's sensors have values

(IR-1, IR-2, IR-3, IR-4) = (25, 50, 200, 180)

1)	0.902	TURN = 0	MOVE = -9.02
2)	0.8	TURN = -72	MOVE = 0.0
3)	0.09	TURN = 8.1	MOVE = 0.0
4)	$0.098 \times 0.8 \times 0.09 = 0.007056$	TURN = 0	MOVE = 0.03528
5)	$0.098 \times 0.2 \times 0.91 = 0.017836$	TURN = 0	MOVE = 0.17836
6)	0.22	TURN = 0	MOVE = 1.1
TOTAL		TURN = -63.9	MOVE = -7.70636

(IR-1, IR-2, IR-3, IR-4) = (200, 10, 20, 80)

1)	0.22	TURN = 0.0	MOVE = -2.2
2)	1.0	TURN = -90	MOVE = 0.0
3)	0.64	TURN = 57.6	MOVE = 0.0
4)	$0.78 \times 1.0 \times 0.64 = 0.4992$	TURN = 0	MOVE = 2.496
5)	$0.78 \times 0.0 \times 0.36 = 0.0$	TURN = 0	MOVE = 0.0
6)	0.922	TURN = 0	MOVE = 4.61
TOTAL		TURN = -32.4	MOVE = 4.906

(IR-1, IR-2, IR-3, IR-4) = (180, 10, 150, 10)

1)	0.3	TURN = 0	MOVE = -9.02
2)	1.0	TURN = -90	MOVE = 0.0
3)	1.0	TURN = 90	MOVE = 0.0
4)	$0.7 \times 1.0 \times 1.0 = 0.7$	TURN = 0	MOVE = 0.03528
5)	$0.7 \times 0.0 \times 0.0 = 0.0$	TURN = 0	MOVE = 0.0
6)	0.41	TURN = 0	MOVE = 2.05
TOTAL		TURN = 0	MOVE = 2.55

(IR-1, IR-2, IR-3, IR-4) = (5, 5, 120, 5)

1)	0.98	TURN = 0	MOVE = -9.8
2)	1.0	TURN = -90	MOVE = 0.0
3)	1.0	TURN = 90	MOVE = 0.0
4)	$0.02 \times 1.0 \times 1.0 = 0.007056$	TURN = 0	MOVE = 0.1
5)	$0.02 \times 0.0 \times 0.0 = 0.017836$	TURN = 0	MOVE = 0.0
6)	0.53	TURN = 0	MOVE = 2.65
TOTAL		TURN = 0	MOVE = -7.05